NAVYA SYAL

CYBER SECURITY STUDENT

CONTACT

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21 - B, Pocket - 1, Mayur Vihar Phase - 1, Delhi - 110091

LinkedIn:
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SOFT SKILLS

- Teamwork
- Time Management
- Leadership
- Effective Communication
- Critical Thinking
- Problem Solving

TECH SKILLS

- Python
- SQL, MySQL
- Shell Scripting
- JavaScript
- HTML
- Linux
- MERN Stack

LANGUAGES

- English (Fluent)
- Hindi (Fluent)

PROFILE

Pre - Final Year Cybersecurity undergraduate with a strong foundation in network security, Linux, Python, and database systems, complemented by hands-on experience in frontend development using React and HTML. Experienced in building responsive, user-centric web interfaces while understanding the security considerations behind modern web applications.
Actively developing expertise in SOC operations, threat detection, log analysis, and incident response through academic coursework and self-driven projects. Passionate about safeguarding digital assets and bridging the gap between secure development and defensive security practices. Brings analytical thinking, problem-solving ability, and a continuous learning mindset to technical challenges.
Currently preparing for CompTIA Security+ and seeking a 6–12 month cybersecurity or SOC internship where I can contribute both application-level understanding and security-focused technical skills in real-world environments.

EDUCATION

NIIT University, Neemrana, Rajasthan — 2023 - Present
Bachelor of Technology - BTech in Cyber Security

Somerville School, Vasundhara Enclave — 2009 - 2023
High School Diploma

PROJECTS

MULTIVARIATE OPTIMIZATION — SEPTEMBER 2023 NOVEMBER 2023

·Implemented Lagrange Multipliers using SciPy for optimization in energy distribution systems.

4-ADJACENCY IN IMAGE PROCESSING — FEBRUARY 2024 - MAY 2024

Used OpenCV and NumPy to implement 4-adjacency logic for pixel connectivity in grayscale images.

SMART FITNESS TRACKER — FEBRUARY 2024 - DECEMBER 2024

·Designed a cross-platform fitness tracker to monitor health metrics using IoT integrations and Thingspeak API. Developed mobile interface with Android Studio.

CAREER COUNSELING MANAGEMENT SYSTEM — FEBRUARY 2025 - MAY 2025

Created a robust database management system to handle psychometric evaluations and generate career recommendations.

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DEVELOPMENT AND DIGITAL TRANSFORMATION OF IOT-ENABLED HYDRAULIC RUBBER MOLDING MACHINE

OCTOBER 2025 - NOVEMBER 2025

Designed and integrated IoT sensors to collect real-time machine data (pressure, temperature, vibration, cycle time) and built a secure cloud-based pipeline for analytics and monitoring. Developed real-time dashboards to track machine health and production metrics, optimizing cycle time and reducing downtime to improve operational efficiency.

QUANTUM ANALYSIS OF AES USING SHANNON'S THEORY

SEPTEMBER 2025 - NOVEMBER 2025

Analyzed AES-128, AES-192, and AES-256 using Shannon's information theory by evaluating entropy, redundancy, and unicity distance to assess classical cryptographic strength. Examined quantum attack feasibility using Grover's algorithm, compared quantum resource requirements with Shannon-based security limits, and evaluated AES resilience in the post-quantum threat landscape.

ATDAM: ADAPTIVE THREAT DETECTION AND AUTOMATED MITIGATION FRAMEWORK (RESEARCH STUDY)

SEPTEMBER 2025 - NOVEMBER 2025

Co-authored a theoretical cybersecurity framework integrating signature-based, anomaly-based, and hybrid detection with automated mitigation to reduce attacker dwell time.
Designed architecture, severity scoring logic, adaptive thresholding, and response workflows to enable real-time threat detection and proportional automated defense.

EXPLORING THE MAZE: A COMPARATIVE STUDY OF PATH-FINDING ALGORITHMS FOR PAC-MAN GAME

SEPTEMBER 2025 - NOVEMBER 2025

Conducted a comparative analysis of DFS, BFS, UCS, IDDFS, and A* algorithms in a PAC-Man game environment to evaluate path-finding efficiency across different maze sizes. Implemented heuristic-based A* using priority queues and cost functions, and determined that A* consistently produced optimal paths with fewer node expansions compared to uninformed search methods.

LEADERSHIP AND CAMPUS ENGAGEMENT

- President, Agraga - The Management Society of NIIT University
- ·Member, Student Affairs Committee (SAC) – Event Coordination, Student Welfare